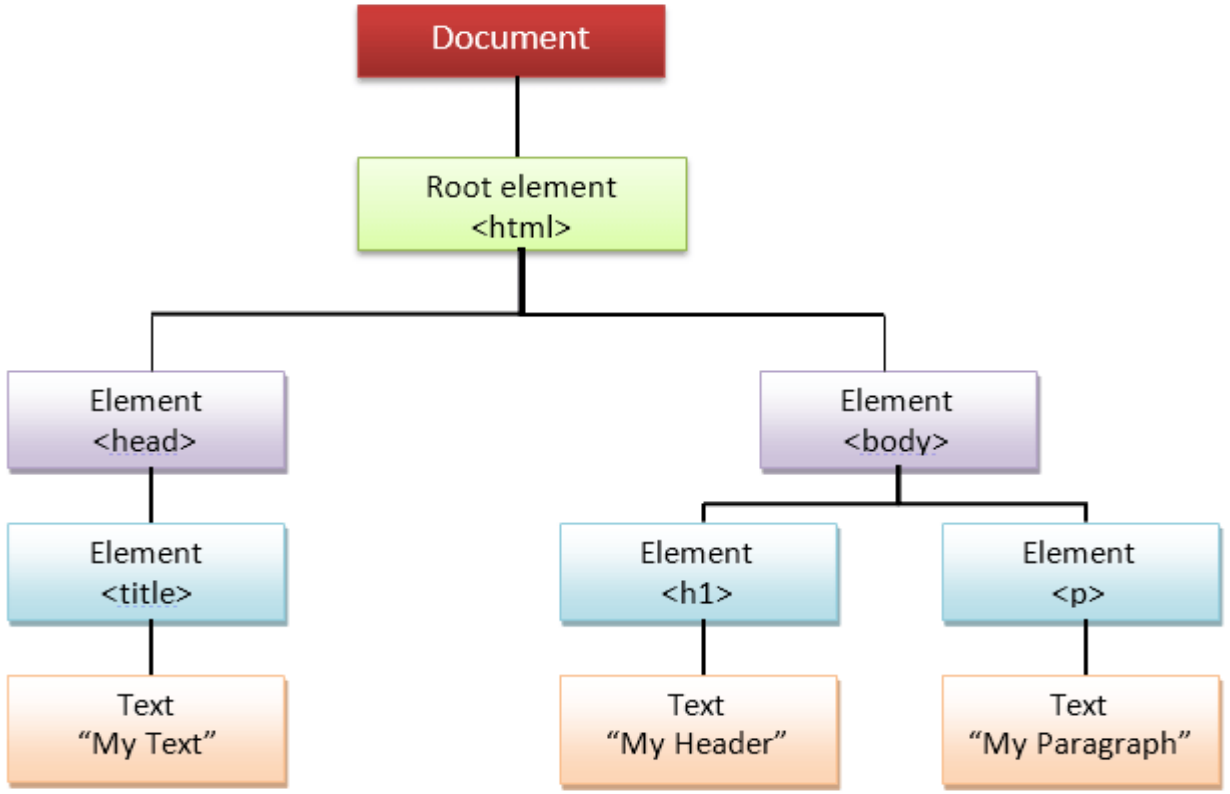
Getting Started With DOM

SUMMARY

1- What is the DOM?
2- How to access the DOM?
3- How to change the html elements?
4- DOM events
5- Node Relationships

1- What is the DOM?

Document Object Model is a programming interface representing the document to enable developers to manipulate the html elements.




```
1   <!DOCTYPE html>
2   <html>
3   <head>
4       <title>My Text</title>
5   </head>
6   <body>
7       <h1>My Header</h1>
8       <p>My Paragraph</p>
9   </body>
10  </html>
```

1- What is the DOM?

Javascript can access the DOM to manipulate the elements and styles of our website:

- JavaScript can add new HTML elements and attributes
- JavaScript can change the HTML elements and attributes in the page
- JavaScript can change the CSS styles in the page
- JavaScript can remove existing HTML elements and attributes
- JavaScript can react to all existing HTML events in the page
- JavaScript can create new HTML events in the page

2- How to access the DOM?

1- To access html elements we should access the document object: " document. "

2- We use methods to select elements:

- document.getElementById() → returns the element that match the specific id
- document.getElementsByClassName() → returns an html collection containing all elements with that class.
- document.getElementsByTagName → returns an html collection containing all elements with that tag name.
- document.querySelector() → returns the first element (with its complete tag) that matchs the selector, we can use any of the css selectors (tag name, id, class) ("tagName", ".class", "#id").
- document.querySelectorAll() → returns a node List containing a list of all elements that match our selector (any selector too).

2- How to access the DOM?

Other methods:

- document.documentElement → returns the entire html element.
- document.images → returns an html collection of all images.
- document.head → returns the head element.
- document.lastModified → returns the date and time the document was updated.

3- How to change the html elements?

Create a button element:

```
//creating an element:
var btn = document.createElement('button');
btn.innerHTML = 'add';
myParagraph.appendChild(btn);
```

Changing the content of the tag:

```
//change the content of the button:
var prg = document.querySelector("button").innerHTML= "delete";
```

3- How to change the html elements?

Add id attribute to the button:

```
//adding id attribute
var btn = document.querySelector("button");
btn.setAttribute('id', 'myBtn');
console.log(btn);
```

Change the value of the attribute:

```
//changing the id value
btn.id = 'bouton';
```

3- How to change the html elements?

Change the button style:

```
//adding style to the h3:
var heading = document.querySelector(".card > h3");
heading.style.color= 'blue'
heading.style.fontSize = '40px';
```

Remove element:

```
//remove an element
var para = document.getElementById('paragraph');
para.remove();
```

3- How to change the html elements?

Remove Child element:

```
//remove child element:
var el = document.getElementById('container').lastElementChild;
el.parentNode.removeChild(el)
```

Replace element:

```
//replace child element:
var list = document.querySelector('ul');
var childList = document.createElement('li');
childList.textContent = "first list";
list.replaceChild(childList, list.childNodes[0]);
```

3- How to change the html elements?

Writing directly into the html output stream:

```
//Writing directly in the html stream
document.write('<header><h1>DOM WS</h1><ul><li>first item</li><li>second item</li></ul></header>')
```

4- DOM Events

Events are actions that happen in the system, which the system tells you about so you can respond to them.

HTML DOM allows JavaScript to react to HTML events. Here are some of these events:

- onclick
- mouse over
- mouse move
- page load

4- DOM Events

Defining events directly in our HTML code using attributes:

```
<button onclick="innerHTML = 'Hello!'">Click me!</button>
```

We can also call functions when an event is triggered:

```
<button onclick="changeText()">Click me!</button>
```

We assign the event to our JavaScript code:

```
//changing the text on click
var changeBtn = document.getElementById("btn").onclick = changeText();
```

5- Event Listener

- AddEventListener() method attaches an event handler to the specified element.

The method attaches an event handler to an element without overwriting existing event handlers.

You can add many event handlers to one element.

You can add many event handlers of the same type to one element, i.e two "click" events.

- removeEventListener() method removes event handlers that have been attached with the addEventListener() method.

6- Node Relationship

- The nodes in the DOM Document have a hierarchical relationship to each other. This means that the nodes are structured like a tree.
- We use the terms parent, sibling, and child to describe the relationship between nodes.
- The top node is called the root and is the only node that has no parent.
- The root in a normal HTML document is the <html/> tag because it has no parent and is the top tag of the document.

parentNode, childNodes, firstElementChild, lastElementChild, nextSibling